

Hives

(Urticaria; Wheals)

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Hives are red, itchy, slightly elevated swellings. The swelling is caused by the release of chemicals (such as histamine) from mast cells in the skin, which cause fluid to leak out of small blood vessels temporarily. Itching may be severe. Hives have clearly defined borders and may have a pale center. Typically, crops of hives come and go. One hive may remain for several hours, then disappear, and later, another may appear elsewhere. After the hive disappears, the skin usually looks completely normal.

(See also [Itching](#).)

Hives



Hide Details

Hives are raised, itchy, and reddish spots on the skin.

Photo provided by Thomas Habif, MD.

Angioedema

Hives may occur with [angioedema](#), which, like hives, involves swelling. However, the swelling of angioedema is under the skin rather than on its surface. Sometimes angioedema affects the face, lips, throat, tongue, and airways. It can be life threatening if the swelling interferes with breathing.

Causes of Hives

Hives and angioedema can be [allergic reactions](#).

Hives may occur when certain chemicals are inhaled, consumed, injected, or touched. These chemicals can be in the environment, foods, drugs (including medications), insects, plants, or other sources. They are harmless in most people. But if people are sensitive to them, these chemicals (called triggers or allergens) can cause an allergic reaction. That is, the immune system overreacts to the chemicals.

However, most of the time, hives are not part of an allergic reaction and an allergen (the cause of the allergic reaction) cannot be identified. For example, they may result from [autoimmune disorders](#). In these disorders, the immune system malfunctions, misinterpreting the body's own tissues as foreign and attacking them. Also, some medications cause hives directly without triggering an allergic reaction. Some physical stimuli (such as heat, cold, pressure, friction, or sunlight) may cause hives for reasons that are not well understood.

Hives usually last less than 6 weeks and are classified as acute. If hives last more than 6 weeks, they are classified as chronic.

Acute hives

If a cause can be identified, acute hives are most commonly caused by:

- Allergic reactions (such as foods and food additives, medications, or insect bites)
- Nonallergic reactions (such as medications, physical stimuli, or autoimmune disorders)

Allergic reactions are often triggered by foods, particularly eggs, fish, shellfish, nuts, and fruits; food additives; medications; or insect bites or stings. Eating even a tiny amount of some foods can suddenly cause hives. But with other foods (such as strawberries), an allergic reaction occurs only after a larger amount is eaten. Many medications, particularly antibiotics, may cause hives. Immediate allergic reactions may also occur when a substance comes into direct contact with the skin (such as latex), after an insect bite or sting, or as a reaction to a substance that is inhaled into the lungs or through the nose.

Nonallergic causes of hives include infections, some medications, and some physical stimuli (such as pressure or cold).

In more than half of cases, a specific cause of acute hives cannot be identified.

Chronic hives

Possible identifiable causes of chronic hives are the same as those of acute hives. However, in the vast majority of cases, a cause cannot be identified (is idiopathic). Most cases of chronic hives that have no identifiable cause are thought to be due to an autoimmune reaction that itself has no detectable cause. Nevertheless, all efforts should be made to identify a cause, because elimination of a cause is the best approach to treatment.

Sometimes the cause is easily overlooked, as when people repeatedly consume a food not known to be a trigger, but the food contains a preservative or dye, or an antibiotic (such as penicillin) has been given to a cow which then is present in the milk obtained from that animal. Often, despite the best efforts, the cause remains unidentified.

Chronic hives can last for months or years, then sometimes go away for no apparent reason.

Evaluation of Hives

Not every episode of hives requires immediate evaluation by a doctor. The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

Certain symptoms and characteristics are cause for concern:

- Swelling of the face, lips, throat, tongue, or airways (angioedema)
- Difficulty breathing, including wheezing
- Hives that are deeply colored, that become open sores, or that persist for more than 48 hours
- Fever, swollen lymph nodes, jaundice (yellowing of the skin and eyes), weight loss, and other symptoms of a bodywide (systemic) disorder

When to see a doctor

People should call an ambulance if:

- They have difficulty breathing or are wheezing
- Their throat feels as if it is closing up

People should go to an emergency department or a doctor's office as soon as possible if

- Their symptoms are severe
- They feel progressively weak or light-headed or have severe fever or chills
- They are vomiting or have abdominal pain or diarrhea

People should see a doctor if:

- A bee sting triggers hives or swelling (to obtain advice about treatment if another bee sting occurs)

- They have other symptoms, such as fever, joint pains, weight loss, swollen lymph nodes, or night sweats
- Hives recur without exposure to a trigger
- Symptoms last for more than 2 days

If children have hives that appear suddenly, disappear quickly, and do not recur, an examination by a doctor is usually unnecessary. The cause is usually a viral infection.

What the doctor does

Doctors first ask questions about the symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause and the tests that may need to be done (see table

[Some Causes and Features of Hives](#)).

Doctors ask the person to describe each episode of hives in detail and any other symptoms that occurred (such as [itching](#), difficulty breathing, or [swelling of the face and tongue](#)). They ask about the person's activities before and during the episode and about possible exposure to substances that can trigger allergic reactions, including medications or drugs being taken. The person is also asked about specific symptoms that could suggest a cause (see table [Some Causes and Features of Hives](#)), recent infections, past allergic reactions, and recent travel.

The trigger is not always clear from the history, often because the trigger is something that may have been tolerated previously.

During the physical examination, doctors first check to see whether the lips, tongue, throat, or airways are swollen. If there is swelling, they begin treatment right away. Then doctors note how the hives look, determine which parts of the body are affected, and check for other symptoms that may help confirm the diagnosis. Doctors may use various physical stimuli to see whether they can trigger the hives. For example, they may apply light pressure, heat, or cold to the skin or stroke the skin.

People themselves should not try to trigger their hives, because a severe reaction could occur.



Some Causes and Features of Hives

Cause	Common Features*	Tests
Acute hives (lasting less than 6 weeks)		
Medication and substances such as • <u>Aspirin</u> and other nonsteroidal anti-inflammatory drugs (NSAIDs) • Some opioids • <u>Vancomycin</u> • <u>Succinylcholine</u> (sometimes given before surgery) • Contrast agents (used with imaging tests such as computed tomography) • Possibly any medication or substance, whether prescription, over-the-counter, or herbal, if people are allergic to it	Hives can start within minutes after the drug or medication was taken and can start up to 48 hours later	A doctor's examination
 Foods that trigger an allergic reaction (food allergens), such as peanuts, nuts, fish, shellfish, wheat, eggs, milk, and soybeans	Hives that start within minutes or hours after consumption	A doctor's examination, particularly the medical history Sometimes allergy skin prick testing
	Fever, chills, and fatigue	Tests depending on the suspected